

Batch-Level Predictors of Training Discontinuities under Aligned, Small-Step, Forgetting-Protected Dynamics: A Formal Restatement and Literature Review

TL;DR

- A defensible combined claim is possible but only as a *conditional* (Lipschitz-style) statement, not a categorical one. Under the four-part premise (NTK alignment of the new batch, small task/mode distance, small step size or warmup, and replay/parameter isolation), training dynamics are approximately linearizable around the current parameters; in this regime, one can prove (i) that a few samples from a new mode improve approximation on both the new mode and not-yet-covered but kernel-aligned modes, and contrapositively (ii) that any rapid change in the training trajectory — loss spike, NTK eigenspectrum reorganization, sharpness jump, or escape from the linearized basin — must be *attributable* to a batch that violates at least one conjunct. The claim is therefore a *characterization-by-violation*, not a positive prediction.
- Among the candidate batch-level scores, only a small subset is genuinely *forward-looking* (predictive) rather than post-hoc (attributive) under formal theory. Predictive: (a) batch sharpness / directional curvature along the batch gradient (Andrejev & Beneventano 2024 preprint, building on Cohen et al. 2021), (b) the alignment between the batch tangent features and the current empirical NTK eigenbasis (Atanasov-Bordelon-Pehlevan 2022; Doan et al. 2021), and, with caveats, (c) the batch gradient norm above an adaptive threshold (Kumar et al.'s ZClip preprint 2025; Rosenfeld & Risteski ICLR 2024 mechanism). Attributive only: influence functions (Koh & Liang 2017), TracIn / TRAK, RHO-LOSS, Fisher trace; these are fundamentally retrospective or contingent on a fully trained reference model. (arxiv)
- A consensus *positive* predictive claim is not yet supported by the literature. Influence-function-style scores are demonstrably fragile in non-convex regimes (Basu, Pope & Feizi ICLR 2021) and RHO-LOSS predicts informativeness, not instability. The strongest current predictor — the joint event "batch sharpness $\approx 2/\eta$ AND batch gradient direction has large component on the top Hessian/NTK eigenvector AND batch is far from the replay buffer in feature space" — is theoretically motivated by edge-of-stability and silent-alignment results but has only been empirically validated in toy and mid-scale settings; LLM-scale evidence is presently from preprints.

Key Findings

1. **The four-conjunct premise places training in a regime where the empirical NTK is approximately stationary in eigenstructure.** This is precisely the regime in which Atanasov, Bordelon & Pehlevan (ICLR 2022, "Silent Alignment Effect") and Geiger et al. (J. Stat. Mech. 2020) prove the dynamics are governed by kernel gradient descent with a (near-)fixed kernel. Inside this regime the function-space update is linear and one can give *generalization* statements for points within the kernel's RKHS span. Outside it, the dynamics become nonlinear and the NTK eigenspectrum can rotate and rescale within a single update — the precondition for loss spikes, catapults, and grokking-type transitions (Kumar, Bordelon, Gershman & Pehlevan, ICLR 2024; Lewkowycz et al. 2020 preprint).
2. **The original informal points 4 and 5 are mutually consistent only when reformulated as a single conditional.** Point 4 (few samples from a new mode help both that mode and uncovered ones) holds *iff* the new batch lies in or near the existing NTK eigenbasis; point 5 (generalization fails on high-surprisal data) holds because high-surprisal batches are precisely those whose tangent features are not in the span of the current NTK top-eigenvectors. The two informal claims are thus the positive and negative branches of a single Lipschitz/perturbation theorem about kernel gradient descent. They cannot both hold *unconditionally* — that is the impossibility argument: high-surprisal but kernel-aligned data does exist (e.g., long-tailed but in-distribution token sequences), so point 5 must be qualified to "data with high entropy *and* low NTK alignment with current features."
3. **Phase transitions/loss spikes/NTK eigenspectrum shifts can be *attributed* to single (or low-cardinality) batches under the premise,** by a contrapositive argument. If the four-conjunct premise holds, the linearized dynamics predict loss decrease at rate proportional to the projection of the residual onto the NTK eigenbasis (Bordelon, Canatar & Pehlevan, ICML 2020). Any deviation from this prediction — measured as a positive residual in the linearized loss model — must originate in a batch that violates at least one conjunct (most often: lost NTK alignment, exceeding $2/\eta$ in batch curvature, or distribution far from replay buffer).
4. **Among forward-looking batch-level scores, three classes have formal grounding:**
 - **Batch sharpness / directional curvature.** Cohen et al. (ICLR 2021) showed full-batch sharpness rises to and hovers at $2/\eta$; Andreyev & Beneventano (preprint 2024, arXiv:2412.20553) generalize this to SGD, showing *batch sharpness* — the expected directional curvature of mini-batch Hessians along their corresponding stochastic gradients — is what stabilizes at $2/\eta$. A batch whose sharpness is significantly larger than $2/\eta$ in advance is a formally predictive instability indicator. (Sitanchen +2 + 2)
 - **Batch-NTK alignment (cosine of batch tangent features with top eigenvectors of empirical NTK).** Atanasov, Bordelon & Pehlevan (2022) and Shan & Bordelon (preprint 2021) show NTK alignment with the target predicts training-time eigenstructure stability. Doan et al. (AISTATS 2021) formalize task similarity through the NTK overlap matrix and show forgetting (i.e., a discontinuity in the trajectory)

increases monotonically with mismatch. (arxiv)

- **Batch gradient norm above an adaptive threshold.** Empirically, ZClip (Kumar et al. preprint 2024, arXiv:2504.02507) and AdaGC (preprint 2025, arXiv:2502.11034) detect spike-precursor gradient norms via z-score anomaly detection. This is predictive only relative to a recent baseline; the theoretical foundation is the Rosenfeld & Risteski (ICLR 2024) "opposing signals" mechanism in which heavy-tailed outliers cause oscillations that culminate in spikes. (arxiv + 2)

5. **Influence functions, TracIn, TRAK, and RHO-LOSS are *attributive*, not predictive, of training-trajectory discontinuities.** Influence functions (Koh & Liang, ICML 2017) estimate the effect of a training point on a *test-time* prediction at convergence; Basu, Pope & Feizi (ICLR 2021) show they are fragile for deep, non-convex networks. TracIn (Pruthi et al. NeurIPS 2020) requires checkpoints already produced. TRAK (Park et al. ICML 2023) requires an ensemble of trained models. RHO-LOSS (Mindermann et al. ICML 2022) selects points by their reducible holdout loss against a *fully trained* reference model — it predicts informativeness for final loss, not instability of the trajectory. None are batch-level predictive indicators of phase transitions. (Google Scholar + 5)
6. **Hessian sharpness and Fisher trace are predictive but coarse.** They predict proximity to the edge of stability (Cohen et al. 2021, Damian, Nichani & Lee preprint 2022, ICLR 2023) but not which *future batch* will tip the trajectory; they are model-state quantities, not batch-state quantities. Combining sharpness with batch-level directional curvature (i.e., batch sharpness) recovers a batch-localized predictor.
7. **The replay/parameter-isolation conjunct is the operational reason the contrapositive attribution works.** Kirkpatrick et al. (PNAS 2017, EWC) penalizes movement in directions of high Fisher information; replay-buffer methods provide the data-side analog. Information-theoretic generalization bounds for replay-based CL (Wen et al. preprint 2025, arXiv:2507.12043) and Hiratani (NeurIPS 2024 preprint, arXiv:2405.20236) show that *with* replay/EWC the trajectory cannot deviate far from the multi-task minimum — so any deviation localizes to specific outlier batches. Without replay, trajectory discontinuities are generic, not batch-attributable. Dohare et al. (Nature 2024) further document loss of plasticity that is gradual and parameter-wide, *not* batch-attributable, providing a complementary failure mode. (PNAS) (Nature)

Details

1. The Restated Combined Claim (rigorous, semi-formal)

Let $f_\theta: \mathcal{X} \rightarrow \mathbb{R}^k$ be a neural network and Θ_t the empirical NTK at training step t with eigendecomposition $\Theta_t = \sum_i \lambda_i(t) v_i(t) v_i(t)^\top$. Let B_t be the batch presented at step t , with empirical loss ℓ_{B_t} and gradient $g_t = \nabla_\theta \ell_{B_t}(\theta_t)$. Define:

- **(A) Feature-reuse / batch-NTK alignment:** $A_t := \cos(g_t, V_k^\wedge(t) g_t) \geq 1 - \varepsilon_A$, where $V_k^\wedge(t)$ is the projector onto the top-k eigenvectors of Θ_t and ε_A is small.
- **(D) Task/mode distance:** $d_{W2}(p_{B_t}, p_{\text{replay}}) \leq \varepsilon_D$ for some Wasserstein metric on the joint feature distribution (or, equivalently, the NTK overlap matrix between B_t and the replay buffer has Frobenius norm $\geq 1 - \varepsilon_D$, after Doan et al. 2021).
- **(η) Small step size / warmup:** $\eta_t \cdot S_t < 2$ where S_t is batch sharpness in the sense of Andreyev & Beneventano (2024, preprint), or $\eta_t \leq \eta_{\max}(t)$ under a warmup schedule satisfying Gilmer et al.'s (NeurIPS 2022 OPT workshop / preprint arXiv:2110.04369) curvature condition.
- **(R) Replay or parameter isolation:** Either every step also samples B'_t from a replay buffer $\mathcal{R}$ with $\|B'_t\| \geq m$, or the gradient g_t is composed with a projection P_t away from the row-space of past task Fisher matrices (EWC; Kirkpatrick et al. 2017).

Combined claim (Conditional Linearization Claim, CLC). *Under $(A) \wedge (D) \wedge (\eta) \wedge (R)$, there exist constants $C_1, C_2 > 0$ (depending on width, depth, parameterization, and $\varepsilon_A, \varepsilon_D$) such that for all training steps in a horizon $T \leq T(C_1, C_2)$:*

(i) **Positive transfer (formalizes informal point 4):** for any held-out distribution q whose support lies in the span of the top-k NTK eigenvectors $v_i^\wedge(t)$ (i.e., q is "kernel-aligned to the current basis"), the change in expected loss ΔL_q satisfies $\Delta L_q \leq -\eta_t \cdot \langle \nabla L_q, \Pi_k g_t \rangle + C_1 \eta_t^2 \|g_t\|^2$, where Π_k is the projector on the top-k NTK eigenspace. Hence even if q is a mode under-covered by the training data so far, a single (or few) batch-aligned sample(s) yields a kernel-regression-style approximation improvement bounded by the same kernel-residual quantity that controls improvement on the new mode itself (Bordelon, Canatar & Pehlevan, ICML 2020; Atanasov-Bordelon-Pehlevan 2022).

(ii) **Failure under surprisal (formalizes informal point 5):** for any batch B with high "kernel-residual surprisal" $\sigma_{\text{NTK}}(B) := \|(I - V_k) \phi_t(B)\| / \|\phi_t(B)\|$ where ϕ_t is the tangent-feature map at θ_t , the linearized predictor incurs error growing at rate $\Omega(\sigma_{\text{NTK}}(B))$, and the gradient step on B drives the dynamics out of the linearization basin if simultaneously $\sigma_{\text{NTK}}(B) > \tau$ and the batch's directional curvature exceeds $2/\eta_t$.

(iii) **Attribution of trajectory discontinuities (the synthesis):** any phase transition observable in {loss spike, gradient-norm spike, top-NTK-eigenvalue rotation, sharpness jump above $2/\eta_t$, escape from linearly connected basin in the sense of Fort et al. NeurIPS 2020} that occurs at step t^* must be attributable to a batch $B_{\{t^*\}}$ (or a low-cardinality window of batches) that violates at least one of (A), (D), (η). Conjunct (R) is necessary to rule out the "gradual drift" failure mode of Dohare et al. (Nature 2024).

Why the unconditional version is impossible. Without conjunct (A) the silent-alignment guarantee fails (Atanasov et al. 2022, §4: anisotropic data delays/breaks alignment); without (η) progressive sharpening alone causes discontinuities even on i.i.d. data (Cohen et al. 2021); without (D) the NTK overlap matrix can be small even with kernel-aligned features (Doan et al. 2021); and without (R) loss of plasticity creates trajectory discontinuities that *are not localizable to specific batches* (Dohare et al. 2024). The four conjuncts are minimal in this sense.

The CLC is consistent with the literature in the form just stated; it would *not* be consistent if it claimed (a) that few-shot positive transfer extends to off-basis modes (contradicts NTK spectral-bias bounds), or (b) that all loss spikes are batch-attributable (contradicts the optimizer/numerical-precision spike sources documented in AdaGC and ZClip preprints).

2. Literature supporting each conjunct

Conjunct (A) — Feature reuse / NTK alignment. Atanasov, Bordelon & Pehlevan (ICLR 2022) prove that for homogeneous nets with small initialization on whitened data, the NTK first evolves in eigenstructure while the loss is still high (the "silent alignment" phase) and then only in scale — converting feature learning into kernel regression with the *aligned* kernel. Their §3-§4 prove this for two-layer linear nets and demonstrate it numerically for nonlinear and multi-output cases. Geiger, Spigler, Jacot & Wyart (J. Stat. Mech. 2020) characterize the lazy-vs-feature crossover via the output-scaling parameter α ; in the lazy phase the NTK is approximately frozen, providing the analytic grounds for premise (A). Kumar, Bordelon, Gershman & Pehlevan (ICLR 2024, "Grokking as the Transition from Lazy to Rich") show that delayed-feature-learning (grokking) is precisely the failure mode when (A) is violated initially and only later restored: networks that "begin in the lazy regime" (their condition (3)) then transition to rich training, with the timing controlled by NTK-task alignment ϵ . Shan & Bordelon (preprint 2021) develop a theory of NTK alignment showing it accelerates training and is governed by data and architecture; Fort, Dziugaite, Paul, Kharaghani, Roy & Ganguli (NeurIPS 2020) document a rapid 2-3 epoch transient in which the NTK changes and after which it stabilizes — i.e., empirically locating when premise (A) becomes valid. (arXiv + 10)

Conjunct (D) — Task / mode distance. Doan, Bennani, Mazouze, Rabusseau & Alquier (AISTATS 2021) introduce the *NTK overlap matrix* as the formal measure of task similarity in the NTK regime; their main theorem expresses catastrophic-forgetting error as a function of this overlap. Hiratani (NeurIPS 2024 preprint, arXiv:2405.20236) disentangles task similarity at the level of features vs. readouts in linear teacher-student models and proves that high feature similarity coupled with low readout similarity is catastrophic for both transfer and retention. Lee, Mannelli, Clopath, Goldt & Saxe ("Continual Learning in the Teacher-Student Setup", ICML 2021 / arXiv:2107.04384) show that *intermediate* similarity yields the largest forgetting — meaning premise (D) cannot be replaced by "small distance is always good"; it specifically requires alignment in both feature and readout subspaces. (arxiv + 3)

Conjunct (η) — Small step / warmup. Cohen, Kaur, Li, Kolter & Talwalkar (ICLR 2021) document the edge-of-stability phenomenon: full-batch GD on neural objectives operates with sharpness hovering just above $2/\eta$, with non-monotone short-timescale loss but long-term decrease. Damian, Nichani & Lee (ICLR 2023, "Self-Stabilization") explain this via third-order self-stabilization. Andreyev & Beneventano (preprint 2024, arXiv:2412.20553) extend to SGD with the Edge of Stochastic Stability and define batch sharpness. Why warmup matters: Kalra, Barkeshli et al. (preprint, arXiv:2406.09405, "Why Warmup the Learning Rate?") show that in the high-LR / no-warmup regime the pre-conditioned sharpness exceeds $2/\eta$ early, causing catapults and training failures; warmup allows pre-conditioned sharpness to decrease naturally below threshold. Goyal et al. (2017, "Accurate, Large Minibatch SGD") established empirically that linear warmup is necessary for large batch training (loss diverges otherwise). Gilmer et al. ("A Loss Curvature Perspective on Training Instability", NeurIPS 2022 preprint) provides the unifying picture: model/HP choices that succeed are precisely those that route the early trajectory through low-curvature regions. (arXiv + 8)

Conjunct (R) — Replay / parameter isolation. Kirkpatrick et al. (PNAS 2017, EWC) provides the canonical parameter-isolation method via diagonal-Fisher-weighted L2 penalty; Huszár (2017 note) clarifies the Bayesian-Laplace interpretation and flags the multi-task quadratic-penalty inconsistency. Dohare, Hernandez-Garcia, Lan, Rahman, Mahmood & Sutton (Nature 2024) demonstrate that without sustained diversity injection (their continual backpropagation), standard methods *gradually* lose plasticity — a trajectory failure mode that is *not* batch-attributable, motivating why (R) is needed for the attribution side of the CLC. Theoretical analyses of replay: Wen et al. (preprint 2025, arXiv:2507.12043) derive information-theoretic generalization bounds for replay-based CL; Lanzillotta et al. (preprint 2025, "Heads collapse, features stay") show small replay buffers preserve feature separability but not head-level alignment, providing a quantitative theory of what replay accomplishes; Banayeeanzade et al. (preprint 2024, arXiv:2408.16939) give overparameterized analyses linking replay buffer size to forgetting and generalization in multi-task and continual learners. (PNAS + 5)

3. Predictive vs. attributive batch-level quantifications

I evaluate each candidate on three axes: (P) is the signal computable *before* applying the batch (forward-looking), (T) is there formal theory supporting its predictive value for trajectory discontinuities, and (C) is the supporting literature contested.

Score	(P) Forward-looking?	(T) Theoretical support?	(C) Contested?	Verdict
Batch gradient norm $\ g_t\$	Yes (computable before update from current θ_t and B_t)	Partial: Rosenfeld & Risteski (ICLR 2024) provide a mechanism (opposing-signal outliers); ZClip/AdaGC (preprints 2024-25) provide adaptive thresholds	Mildly. GraNd-at-init for <i>data pruning</i> was contested (Kirsch 2023 preprint reproducibility note); GraNd later in training is robust. OpenReview For <i>spike prediction</i> the empirical evidence is strong but mostly from preprints.	Predictive (weak): useful as anomaly detector relative to running mean/variance, not as absolute threshold.
Batch-NTK alignment $\cos(\phi_t(B), V_k(t))$	Yes (requires per-batch Jacobian and an estimate of top NTK eigenvectors)	Strong: Doan et al. (AISTATS 2021) for forgetting; Atanasov-Bordelon-Pehlevan (ICLR 2022) for trajectory linearization; Bordelon-Canatar-Pehlevan (ICML 2020) for spectral generalization	Not contested theoretically; computational cost is the practical issue	Predictive (strong, expensive): the cleanest formal candidate.
Batch entropy / model surprisal	Yes (forward pass only)	Weak/indirect: high surprisal <i>correlates with</i> high $\sigma_{\text{NTK}(B)}$ only when features are aligned; otherwise it conflates noise with novelty	Not directly contested for spike prediction, but RHO-LOSS work shows raw loss/surprisal alone is dominated by noise (Mindermann et al. ICML 2022)	Predictive (very weak alone): must be combined with NTK alignment to disambiguate noise from off-basis novelty.
Reducible loss (RHO-LOSS, Mindermann ICML 2022)	Yes (requires only a pretrained "irreducible" reference)	Bayesian-decision-theoretic for <i>holdout-loss reduction</i> ; not for instability	Mild — RHO-LOSS is correlated with sequence length in language modeling (DavIR preprint 2023)	Not predictive of training discontinuities: designed for informativeness, orthogonal to

Influence functions (Koh & Liang ICML 2017)	No (defined retrospectively for test predictions of trained model)	Convex theory only; Bae et al. and Saunshi et al. give partial deep-learning theory	Yes — contested for deep nets: Basu, Pope & Feizi (ICLR 2021, "Influence Functions in Deep Learning Are Fragile") show estimates are erroneous for deep architectures (arxiv) (OpenReview)	Attributive only; fragile.
TracIn / TRAK	TracIn requires checkpoints; (NeurIPS) TRAK requires model ensemble — both retrospective	TRAK has theoretical grounding via NTK approximation (Park et al. ICML 2023; Tong et al. preprint 2026) (arxiv)	TRAK's accuracy guarantees are conditional and recently characterized (Tong et al. preprint) (arxiv)	Attributive.
Hessian sharpness $\lambda_{\max}(\nabla^2 L)$	Model-state, not batch-state — predicts proximity to EoS only	Strong theory (Cohen et al. ICLR 2021; Agarwala, Pedregosa & Pennington ICML 2023; Damian-Nichani-Lee ICLR 2023)	Theory robust; some refinements ongoing	Predictive but coarse: doesn't localize to a specific batch.
Batch sharpness (mini-batch directional curvature along $\mathbf{g}_t$)	Yes (per-batch quantity, computable via Hessian-vector product)	Andreyev & Beneventano (preprint 2024) show this is what stabilizes at $2/\eta$ under SGD; (arXiv) Lowell & Kastner (preprint 2024, arXiv:2406.00127) link to layerwise Jacobian alignment	Preprint stage but follows directly from Cohen et al.	Predictive (strong, batch-localized): strongest specifically batch-level scalar predictor.
Fisher information trace $\text{tr}(\mathbf{F}_B)$	Yes (per-batch)	Approximates expected curvature; Karakida et al. (2018-19), Achille et al.	Diagonal Fisher is a coarse approximation (Hiratani 2024)	Predictive (medium): correlates with sharpness but isn't

		(Critical Learning Periods)	shows full-Fisher matters)	anisotropic enough to flag direction-specific instability.
EL2N / GraNd later-in- training (Paul et al. NeurIPS 2021)	Per-example, per-step	Heuristic; correlates with instance gradient magnitude	GraNd-at-init is contested (Kirsch 2023 preprint); later-in-training EL2N is reproducible OpenReview	Predictive of importance, weakly predictive of instability when combined with norm anomaly detection.
Outlier-with- opposing- signal detector (Rosenfeld- Risteski ICLR 2024)	Requires identifying paired groups by feature- magnitude; partially forward-looking	Strong mechanistic explanation linking outliers → progressive sharpening → spike arxiv	Recent, novel; mechanism well- supported	Predictive (mechanistic): the most causally explanatory account currently.

Synthesis on predictive scores. The literature genuinely supports a *composite* batch-level instability predictor:

$$\text{PredictiveSpikeScore}(\mathbf{B}; \boldsymbol{\theta}_t) := w_1 \cdot \mathbb{1}[\|\mathbf{g}_B\| > \mu_g + \kappa \sigma_g] + w_2 \cdot \mathbb{1}[\hat{s}_B(\boldsymbol{\theta}_t) > 2/\eta_t] + w_3 \cdot \sigma_{\text{NTK}}(\mathbf{B}; \boldsymbol{\theta}_t)$$

where $\hat{s}_B$ is batch sharpness (Andreyev & Beneventano 2024), μ_g, σ_g are running mean/std of recent batch gradient norms (ZClip preprint), and σ_{NTK} is the kernel-residual surprisal of §1. No single component is sufficient and no peer-reviewed paper has rigorously validated this composite, but each component is grounded in published theory plus active preprints.

4. Where the literature is divided (and the impossibility argument)

There **is** a divided literature on three points relevant to the combined claim:

(a) **Cause of progressive sharpening / EoS.** Cohen et al. (2021) presented it as empirical phenomenology; Agarwala et al. (ICML 2023) attribute it to second-order regression structure; Damian, Nichani & Lee (ICLR 2023) attribute it to third-order self-stabilization; Rosenfeld & Risteski (ICLR 2024) attribute it to data outliers with opposing signals. These are not mutually exclusive but emphasize different drivers, which means a fully *causal* batch-level predictor cannot yet be uniquely specified. [arXiv + 3](#)

(b) **Whether grokking-like transitions are necessary or contingent.** Kumar et al. (ICLR 2024) argue lazy-to-rich transition is *sufficient but not necessary* for delayed feature learning; Liu et al., Varma et al., and others propose competing explanations. This affects whether long-horizon discontinuities are batch-attributable at all (CLC clause iii is therefore qualified to local discontinuities, not grokking-scale ones). [arXiv](#)

(c) **Reliability of influence-style attribution.** Koh & Liang (2017) presented influence functions optimistically; Basu, Pope & Feizi (2021) showed they are fragile for deep nets; TRAK and follow-ups partially restore reliability via NTK approximation but only at the cost of ensembles or randomized projections. Anyone proposing influence-based predictive scores must flag this division. [Google Scholar + 2](#)

The impossibility part. A *categorical, unconditional* form of the combined claim — "few new-mode samples always help, high-surprisal data always fails to generalize, and any phase transition is always due to a single batch" — is impossible, for the following reasons:

1. **Counterexample to "high surprisal always fails":** under (A), a kernel-aligned but high-entropy batch (e.g., samples in the long tail of a power-law distribution but spanning the same NTK top eigenvectors) generalizes well. Surprisal and NTK-alignment are independently varying quantities (cf. Bordelon–Canatar–Pehlevan ICML 2020 spectral bias).
2. **Counterexample to "phase transitions are batch-attributable":** Dohare et al. (Nature 2024) document loss-of-plasticity as a *gradual, parameter-wide* discontinuity with no offending batch. So the attribution part of the CLC requires (R) (replay/parameter isolation) precisely to *exclude* this counterexample. [arxiv](#)
3. **Counterexample to "few-shot positive transfer always works":** Lee et al. (ICML 2021) and Hiratani (2024 preprint) show intermediate task similarity yields *worse* outcomes than either high or low similarity — i.e., the relationship between mode distance and transfer is non-monotonic. So the transfer claim must be qualified to the "kernel-aligned" branch. [arxiv](#)
4. **Counterexample to "small step size suffices":** Cohen et al. (2021) document that even at modest η , progressive sharpening drives the system to EoS where short-timescale loss is non-monotone — small-step does not prevent local discontinuities, only their accumulation into divergence. [Sitanchen +2](#)

These four counterexamples are exactly the four conjuncts of the premise. The CLC is therefore the *strongest* combined claim consistent with the literature.

Recommendations

Stage 1 — Implement the cheap, batch-local indicator first. Track per-batch gradient norm $\|g_B\|$ and maintain a windowed mean/std (ZClip-style). Fire a "potential discontinuity" flag when $\|g_B\| > \mu + 3\sigma$. This costs essentially nothing and, per Rosenfeld & Risteski (ICLR 2024) and the LLM-stability preprint literature, captures the dominant empirical spike-precursor. **Threshold to escalate:** if false-positive rate on benign training is below $\sim 1\%$ and recall on real spikes is below $\sim 70\%$, move to Stage 2.

Stage 2 — Add batch sharpness via stochastic Lanczos / Hessian-vector products on each batch (or on a 1-in-N schedule). This is the formally strongest batch-level scalar (Andreyev & Beneventano 2024, building on Cohen et al. 2021). Compute $\hat{s}_B = (g_B^\top H_B g_B) / \|g_B\|^2$ where H_B is the batch Hessian; flag when $\hat{s}_B > c \cdot (2/\eta)$ with $c \approx 0.9$. **Threshold to escalate:** if you observe NTK eigenrotation (loss of feature reuse) without a sharpness flag firing, move to Stage 3.

Stage 3 — Add batch-NTK kernel-residual surprisal $\sigma_{\text{NTK}}(B; \theta_t)$. Compute the projection of batch tangent features $\phi_t(B)$ onto the top-k empirical-NTK eigenvectors estimated from a running buffer of recent batches (or, in the continual-learning case, from the replay buffer). High σ_{NTK} with low gradient norm and low batch sharpness predicts grokking-like *delayed* discontinuities (Kumar et al. ICLR 2024). This is expensive (requires per-batch Jacobian) and should be applied only at checkpointed intervals or for high-stakes runs.

Stage 4 — Use the composite PredictiveSpikeScore for routing decisions. When the score crosses a learned threshold, take *one* of: (i) skip the batch and add it to a quarantine buffer (PaLM-style, per Chowdhery et al. 2022), (ii) clip its gradient adaptively (ZClip), (iii) replay from buffer to restore (D), or (iv) apply a temporary learning-rate reduction (per Gilmer et al. NeurIPS 2022). Choice between (i)–(iv) should be governed by which conjunct of the CLC is being violated (gradient norm $\rightarrow \eta$; alignment loss $\rightarrow A$ or D ; etc.).

Avoid for instability prediction: influence functions, TracIn, RHO-LOSS, raw entropy/surprisal alone. These should only be used for their intended purposes (data attribution, data selection for end-task loss).

Benchmarks for changing recommendations: if a forthcoming peer-reviewed paper (a) replaces batch sharpness with a stronger SGD-localized predictor, (b) demonstrates that influence-function variants survive in deep networks beyond the toy regime tested by Basu, Pope & Feizi (2021), or (c) provides closed-form NTK-residual scores at LLM scale, then promote those into Stages 2–3.

Caveats

- **Preprint flags.** The strongest formal grounding for *batch-localized* prediction (Andreyev & Beneventano 2024 on Edge of Stochastic Stability; ZClip 2024; AdaGC 2025; Rod Flow 2026) is from arXiv preprints, not peer-reviewed papers. Cohen et al. 2021, Atanasov et al. 2022, Kumar et al. 2024, Damian et al. 2023, and Doan et al. 2021 are peer-reviewed (ICLR/AISTATS); Cohen et al. is the load-bearing peer-reviewed citation for stability-side claims. Treat preprint-only claims as provisional.
- **Width / parameterization caveat.** Most NTK-alignment theory (Atanasov-Bordelon-Pehlevan, Geiger et al., Kumar et al.) is exact only in the wide-network limit or for specific parameterizations (μ P, NTK-parameterization). Finite-width corrections are an active area (Bordelon & Pehlevan NeurIPS 2022, 2023). Real LLM training is finite-width and uses Adam, not SGD; the EoS results extend qualitatively (Cohen et al. 2022) but quantitative thresholds require empirical recalibration. (Jmcohen)
- **The CLC is a Lipschitz-style guarantee, not a probability statement.** It says: "if the four conjuncts hold *throughout the horizon*, then properties (i)–(iii) hold." It does not say a single isolated batch satisfying the conjuncts has the properties; the dynamics are path-dependent. The contrapositive (iii) is therefore the most useful and most defensible part — it gives an attribution rule, not a forecast.
- **"Phase transition" is overloaded.** I use it to mean *measurable trajectory discontinuities*: loss spikes, gradient-norm spikes, NTK eigenrotation, sharpness jumps, escapes from linearly connected basins. The CLC does not address Bayesian phase transitions in the Watanabe sense or critical-period transitions in the Achille et al. sense; those would require additional statistical-mechanics machinery.
- **"Low NTK alignment" and "high surprisal" are not the same quantity.** High surprisal under the current model is *necessary but not sufficient* for low NTK alignment. A predictor based only on surprisal will conflate aleatoric noise with off-basis novelty; this is one reason RHO-LOSS, while excellent for its intended purpose, is *not* a spike predictor.
- **The literature on replay theory is young.** Wen et al. (2025) and Lanzillotta et al. (2025) are preprints; the precise formal characterization of "what replay accomplishes for the training trajectory" is still being established. The reliance on conjunct (R) in the impossibility argument is therefore well-motivated but not yet fully formalized in peer-reviewed work.
- **Computational cost.** Per-batch Hessian-vector products and per-batch NTK projections are expensive at LLM scale. The Stage-1 indicator (gradient-norm anomaly) is the only one that is essentially free; everything stronger requires $\geq 1\times$ extra forward/backward passes per checked batch.